Tech Disruptors: AWS on Managing AI Costs and Enterprise ROI

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Anurag Rana (00:00:20)
Hi everyone, welcome to another episode of the Tech Disruptors podcast. My name is Anurag Rana. I'm a technology analyst at Bloomberg Intelligence, part of Bloomberg's research department with 500 analysts and strategists working across all major world markets.

We're delighted to have Chris Hennesey as our guest today. Chris has been with AWS for over five years now, where he is an executive in residence. As an enterprise finance strategist, he works with executives around the globe to share financial management experiences and strategies for how the cloud can help them increase speed, agility, while devoting more of their resources to their customers.

Prior to joining AWS, Chris worked at Capital One for over 21 years, where he held multiple senior technology finance roles. On today's podcast, we will focus on ongoing AI-related investments by enterprises and the return that they are seeing on those investments. Chris, welcome to the podcast.

Chris Hennesey (00:01:17)
So excited to be here.

Anurag Rana (00:01:19)
Perhaps we start with your background and your current role at AWS, please.

Chris Hennesey (00:01:25)
Definitely. Yes, you said it well, in terms of my experience, where I spend spending a lot of time now with customers at AWS is a lot around agentic AI and AI value realization. So, I've had the pleasure of engaging with customers all across the globe, both CIOs, CTOs, but I spend a lot of time with CFOs as well, who are asking a lot of these same type questions.

Anurag Rana (00:01:49)
So, our media friends do a lot of postings around how companies are burning their annual token budgets within months and AI costs are getting out of hand. And tell us a little bit about what's the reality behind the scenes, given that AWS works with most enterprises

around the world.

Chris Hennesey (00:02:08)
Yes, I think there's a lot of stories and I think a lot of these are true. I do a lot of events where there's groups of customers together, and many of them are trying to figure out, one, how do you plan for and budget for expected token costs? But then two, what are the --what's the ways in which you can articulate the value being realized?

So, I think the stories are real. I think a little bit of the opportunity here is it's not that AI is unaffordable. I think for any one of us that consume AI, it feels affordable locally.

When companies scale them, sometimes they're a bit unmanaged, especially as they're an early technology. So, there's a lot of corollaries to, I know in my experience, going through a cloud transformation, as well as obviously through an AI transformation, a big opportunity for governance and guardrails as you go through this journey.

Anurag Rana (00:02:58)
So we talk to a lot of companies, and I think at this point, most of them have given their employees one or two AI tools, whether it's a co-pilot or cloud.

Chris Hennesey (00:03:09)
Yes.

Anurag Rana (00:03:09)
Or -- so where is this overage coming from? Because to be honest, I've been using it for a few years, and I don't think any -- I've never got a notification that I'm above my token limit or anything like is it -- are these the developers? Give us where the who are the problem children in this particular equation.

Chris Hennesey (00:03:28)
Yes. It's a good question. I don't know if there's problem children. But I do know, if you look at where the majority of consumption is today inside of (inaudible -00:03:36) enterprises, the development teams are a large portion with coding assistants.

The beauty, though, of agentic AI is we're seeing a pretty diverse change in who the buyer pool is of who's leveraging them, because there's a lot of spend happening outside of IT across that. So, I think to your question, one, any one of us who pays a subscription fee, I don't think that's usually where you see the overage. Yes, there's some inference costs that's there.

But two, it's when you start leveraging agents in a way and scaling this in terms of especially external customers consuming this, there's a predictability concern that many customers are trying to work through. So, I think it's a balance inside of an organization. One, what are the developers leveraging to support and build applications?

But two, what are non-developers, I'll call them knowledge workers, consuming? And then how are customers also consuming as products and services get deployed on this? So, a lot of this is just an evolving landscape.

And the customers are trying to get used to the new terms around how cloud providers are billing for these services, and then how to predict the growth in some of these services. So, I think it's a bit of immature, early capabilities for customers to build the muscle around how to be well-managed in leveraging this versus anybody doing things they shouldn't be doing. But I'm sure as you're an active user of AI, I always find myself questioning how many prompts I need to write before to get to right outcomes.

What level of context do I need to give it? Sometimes I provide too much context. So, there's a lot of levers customers can do as they go through this.

But I think a lot of this is we're in the early days of how to use this and use it well from a cost management standpoint.

Anurag Rana (00:05:20)
So I have a friend who was a developer and we were chatting and he was saying, hey, I'm getting a little red signal on my device that my agent is running in the background. So, I was very curious about it as how are companies at least monitoring it right now? So, one part you said is right.

All of us have that subscription product right now. And this is not where the overage is coming for. But let's say coding tools and somebody is using that.

So even let's not even get into a sophisticated enterprise, majority of the use case today or the majority of, let's say, Fortune 2000, how are they even monitoring the use of these tokens? What kind of tools they have at their disposal right now?

Chris Hennesey (00:06:04)
Yes, I think there's multiple layers. One, there's some observability at the LLM or model layer that people are leveraging. So, these are kind of purpose built, kind of the front end of understanding how many calls am I making?

How many tokens am I using? What is the cost profile of that? So, I think that's one angle is the LLM providers themselves.

I know cloud providers, including AWS, also provide transparency and the ability to provide governance and oversight around that. I'm also seeing a huge surge of startups. I'm sure you're seeing that as well.

A lot of startups are building capabilities and tools tied to not only cost attribution, alignment down to the developer team or value stream level, but it's an element of looking at anomalies, looking at kind of overall token usage. And then a lot of companies are

considering kind of limits and ceilings, like you mentioned, kind of kill switches around it, not exceeding certain thresholds in terms of those runaway scenarios. I'm sure some of us have.

So, I think there's a really good balance between cloud providers providing tools, the LLMs providing tools, but then especially customers who are using, and in my case, meeting with customers. Most aren't all in on one model type. They're looking at multiple model providers.

So, there's a lot of great companies who are looking holistically across them to create that single pane of glass as well. So, I think there's a lot of options. It's more a matter of what's your configuration and then what are the practices you want to put in place inside of your company?

Anurag Rana (00:07:38)
So this is nothing new. Whenever a new technology comes in, there are certain cases that are odd. And I can take you back to the comment I remembered back, I do not know, I mean, 25 years ago where one of the CIOs was complaining about how tedious an SAP deployment is, or even in the beginning of cloud days, they were like, oh, well, AWS is pay-per-use, but the bills can get out of hand.

So, give us some context as you have seen the world evolve, when you walk into a scenario where you don't know where the end-to-end costs could be in the services world you talk about, you know, trying to build a fixed outcome project where you don't know what the time and materials would be. So, there are so many use cases there. How do you stack token consumption compared to some of these other use cases?

Chris Hennesey (00:08:31)
Yes, I think there's a couple of things to consider. One, I think the pace of change is pretty rapid. I've been through some of the cycles that you've mentioned.

And while at every point we felt like that pace was fast, I think all of us can attest this is moving at a pace that's pretty fast. So, keeping up with that. The other thing is, as I think of like a large ERP that has a little bit front-loaded cost predictable, cloud was consumption-based, but fairly stable unit costs.

I think on AI, the things for me that this feels a little bit more accelerated in terms of volatility is one, the unit cost is evolving quickly. I think, I know you've referenced in a lot of your other materials, and there's a lot of publications out there of how much the deflationary elements are on inference. But two, the usage patterns are so different.

I think that's one thing a lot of customers are struggling with is, how do I predict the level of consumption across the enterprise and plan for that? So, I think this balance of differences, and as you've seen, the same question can cost materially times different depending on the model and the type and the architecture and the prompt that it's done. So, this feels a little bit different in that regard.

But I know from my vantage point, the advice I typically give customers, one, kind of start with the unit economics and the base of this. I think too many customers I meet with, there's a lot of soundbites inside of enterprises, but it's unclear that data is backing them. So, I think starting with some of the specifics and understanding this goes a long way.

I also advise customers a lot on baselining. I'm sure you hear this. All of us have our own soundbites around how AI is transforming and enabling us to be more productive.

But I know many CFOs, when they hear developers are producing 40% more code or they're also wanting it to translate to the P&L in terms of value, and some of this is productivity gains being reinvested locally. So, I think there's a lot of choice for customers as they go through this. But I think starting with some of the baselining, the unit economics on the front end, and then just as you know -- if you eat an elephant one bite at a time, it's biting off elements of this and isolating the components.

I recently was working with a large government institution across the globe, and they wanted to produce a five-year token forecast for their enterprise. And I know you're very knowledgeable in AI, doing that is a very challenging element. So, we had to kind of break it down component by component to begin to understand even what the range of outcomes can be.

And I think range of outcomes is a good way to think about this, because I don't think anybody's going to be highly accurate in a point estimate at this point.

Anurag Rana (00:11:14)
So Chris, like there is obviously there are certain AI startups that are funded by private equity investors, where it may not matter what the initial cost is, but you can't do that for Fortune 2000 publicly traded company. Enterprise IT budgets just don't balloon at that certain rate. So, when you look at and talk to, it doesn't matter what vertical it is, are there any common threads right now where you see some enterprise IT way stage or where a large portion of the funding is going with very little you could say return on investment?

Chris Hennesey (00:11:52)
I think to the Fortune 2000 companies that are out there, most companies I've met with are funding AI as an incremental cost in the near term. Plenty of private equity-backed companies or maybe government institutions, a lot of them are trying to self-fund these efforts because they don't have quite the desire for growing expenses or the capacity to do that. So, I do spend a lot of time around self-funding opportunities, but I think the economics for me as I've looked into this, you need to be viable.

I think Gartner referenced total global IT spend is expected to grow somewhere around 14% year over year this year. But if you looked at the S&P 500 revenue, it's only expected to grow about 8%. So, you need to make sure that if we are investing at this level in IT, that it's actually producing value and resulting in outcomes that are appropriate over the long term.

So, I think time horizon is a big deal through this. I think a lot of this is fairly new and early, but to the same extent, many CFOs that I meet with are yearning for value delivery and looking for the right ways to measure and hold people accountable inside of the organization towards that. So, I think we're in the seam.

I talk a lot about being in the seam here because there is a lot of investment that's going into third-party models themselves. There's a lot of investment also still in a lot of the data infrastructure. As you know, it's the heart and the context that will enable good outcomes.

So, there's also this element of kind of the iceberg where inference and clear costs you can see, but there's a lot of work that's happening inside of enterprises below the water level to ensure they can take full advantage of the capabilities that AI provides.

Anurag Rana (00:13:36)
Tell me one thing. Is it forcing companies to accelerate cloud shift or is it coming at the expense of cloud shift?

Chris Hennesey (00:13:43)
Yes, I don't know. I think most companies who invested in their cloud transformation are seeing big paybacks in the ability to take advantage of AI now. I think for the companies that are still running on-prem, I think this is highlighting the need and the ability to transform and leverage cloud technology.

But I also meet with plenty of customers who are looking at procuring their own infrastructure to run AI for the high-volume use cases. So, I think most organizations are trying to figure out what's best for them and their business model and their industry. But I think you're spot on specifically, and this is highlighting the need to continue to modernize and be prepared for having the nimbleness and the architecture to take advantage of AI.

New models come out every -- it feels like every week, every couple of weeks. So, the investments in architecture are the ones that will pay back through time. The flexibility then is you need to integrate models in and out.

That's where you need to have a little bit, a lot more flexibility and fungibility inside of their technology stack.

Anurag Rana (00:14:45)
So Chris, as part of preparing for this discussion, I was reading your blog and I was a bit surprised that you mentioned that 95% of enterprises still run on the most expensive model. Now, to be honest, I mean, is that true? Because most of the executives of AI-native companies that we talk to, they have a very sophisticated model routing system.

Or it's just that Fortune 2000 is just way behind some of these AI native firms.

Chris Hennesey (00:15:14)
Yes, that 95% was referenced in terms of the Glean's CEO that Arvind, his name's Arvind, he did in late May, early June. From my vantage point, I do think there is a lot of large enterprises that are deepening a relationship with a single model provider and leveraging, again, while there may be a sliding scale of models within that model provider, they're still leveraging a lot of frontier elements. I think to your point, AI-native companies are very different than large enterprise.

But it's accelerated quickly, even since I've written this blog last week. And I see geographic differences as well, as you may, between EMEA in the Europe area, South America, North America. I was in South Africa last week, engaging with customers there.

It's pretty unbelievable how fast things are evolving and companies are adopting multiple models and providers as a way to ensure they're balancing the cost with the value that they're getting. But I would say a lot, maybe 75% of customers that I view with an enterprise level are still leaning with a single model provider and just trying to figure out what their strategy is, what's the approach they should do. And as you know, a lot of this is less about the technology and more around the culture and the people change involved in this, because there's a lot of questions we get from customers on our team about that.

Anurag Rana (00:16:37)
Very interesting. So when you say 75% that focusing on or pivoting towards one model company, is this more you're talking about the AI diffusion within their operations or you're talking about the tools that they are giving their employees, the subscription tools?

Chris Hennesey (00:16:54)
Yes, probably the tools. Obviously, Anthropic comes up quite often and it is a large portion of that 75%, at least for my discussions, where obviously there's a lot of models within -- model types within Anthropic, but many are looking to leverage the best models for the right workloads. But I think, as you likely know, many customers are leveraging Microsoft for their office productivity tools.

So, there's a great integration with Copilot that's there. Obviously, here at AWS, a leading cloud provider, a lot of companies are leveraging Bedrock, which is kind of the platform where which we provide a model choice for customers in that regard. So, I think customers are early in the journey of figuring out their strategy, what's the right tool set, but then ultimately, how do you deploy that tool set inside of the organization?

I would say highly regulated institutions obviously are holding a very high bar. There's some of the more mature, especially in financial services, some of the more mature companies who are taking advantage of AI because they've done so much of that cloud modernization over the last decade. But as you go through this, it's a fine balance to deployment.

I do a lot of work with the office of the CFO, Finance and Accounting. That's one of the top three jobs that have opportunities to leverage this. But most CFOs are still trying to figure out

how should I leverage agentic AI?

What role does it play as an augmentation versus an automation tool? And very early POCs, I have not met any large enterprise that's fully transformed their organization, where some of the examples you mentioned on the AI-native companies, they probably were born in creating AI operations across HR, legal, finance operations. That's just not the case for large enterprises.

They're trying to figure out where should they reimagine these processes.

Anurag Rana (00:18:45)
So it's clear. I'll put this in two different buckets. One is the tools that you get, where your signal goes to the cloud and comes back. I mean, whether it's OpenAI, Gemini, Claude or Copilot, that's a different entity because you do need a frontier model for that because you don't know what the employee will ask or what they're working on.

And let's go to the other side of the equation where you're adding more agentic capabilities through your operations, through the diffusing more capabilities through that. So let's talk about the second half where, in this case, are they matching model to the kind of work that they're doing or it's still a blanket that, hey, I've just licensed it with this company, whether it's OpenAI or Anthropic, and I'll just use all their models, whether it's in HR, customer service, et cetera, et cetera.

Chris Hennesey (00:19:36)
Yes, I think for my engagements, it's very much the latter, which is, hey, we found a partner we want to leverage, say it's Anthropic or OpenAI. We want to lean in in leveraging that. So, there's probably a little less matching model for agentic platform at this point.

I think what you're saying, though, it's going to evolve to that, which is how do we balance, I think of it as a triangle between accuracy, speed, and cost. And I think any one of us, well, if you had no cost consideration, using a frontier model is great and will support your needs. But once you start scaling some of these elements inside of your organization, that's where if you just look at the differences between, say, an open-weight model versus a frontier model, and there's a lot of choice within frontier as well.

It's trying to find that balance, again, between accuracy, speed, and cost. And that level of maturity is just happening, at least in front of the customers that I'm engaging with. They're trying to find ways to optimize.

They're also, I get a lot of questions of the cost comparisons of human and labor cost versus software and AI cost. What's the right ratios? How should we think about this?

I'm sure there's been some leaders out there, especially on the development side, that you should be consuming a lot as a developer in order to get the velocity and the impacts. It's not quite that extreme in terms of that level of consumption on the knowledge worker or

non-developer side. But I think that maturing is happening now in front of my eyes.

So, I don't see many enterprises who have full optimization on everything. Most are trying to leverage intelligent routing for the questions and let the technology solve this. And that is where I know, partnering with AWS, there's a great tool set there embedded in ours around intelligent routing.

Most are trying to build that orchestration layer because one of the big levers in cost management of AI is around caching. So how do we not have the same people in the same organization paying for the same information? Because every time you're putting new inputs into the LLM, there's more of a bill that's coming.

So, there's a lot of that orchestration layer being created in a lot of the customers. That maturing is happening in front of our eyes right now.

Anurag Rana (00:21:56)
So Chris, perhaps just think about one or two customers that are a little bit more mature that have moved on from one frontier model to multiple models. Give us an overview of where are they pivoting to? Is it small language models?

Is it open-source models? Is it open-weight models? What is the next level of evolution in this journey?

Chris Hennesey (00:22:20)
Yes, I think you just gave a good distribution of the type of models that exist in there. And I think they're the mature companies that I engage with. There's an element of being very deliberate about the use and the ways in which they leverage tools and in what way.

So, it's really about what are the jobs to be done and then what are the best approaches to get those jobs done as a part of it. So, they also do one of the things they typically do is they have the guardrails and visibility up front, but also the optimization set up there as well. So, they set up, I would say, that core infrastructure and architecture layers on the front end as they go through that.

They also give the transparency. One of the patterns I'm seeing with a lot of more mature customers is they're aligning a lot of the AI cost directly to where the value is intended to be delivered so that they can actually have full accountability and visibility around telling that full value creation story as a part of it, which is encouraging to see. And I think there's just a traceability that I think is kind of a blind spot for people that don't have it that I'm seeing in that regard.

So, the other thing I'm excited to see in a lot of enterprises of more mature customers is this tends to be a little less IT-led and a little more product or business-led. Obviously, IT is advising and guiding, but I've just seen the level of technical knowledge expand pretty dramatically, especially in a company that runs in a bit more of a product operating model

mindset where it's integrating tech and business really tightly. I think they're fairly mature in leveraging AI platforms, and you could see that integration and understanding and the ability to know one size doesn't fit all.

So, they may experiment with frontier, but they scale through open-weight models or smaller models that will meet the business needs and balance that triangle that I mentioned.

Anurag Rana (00:24:21)
Well, let me go back and go through that set of companies that are pivoting towards one frontier model. And something came to my mind that when Apple first came out with its integration with OpenAI, OpenAI was just so far ahead of everybody else. And now that's obviously not the case.

Now, from our colleague, Mark Gurman, who has done some reporting on it, basically is down the road, Siri is going to be so sophisticated that you can ask Siri to pick whichever model you want down the road. Or for that matter, you can do it in the back end. Are these companies building an infrastructure where the models are fungible?

Or will they have to go back and now redo the whole thing again? Let's say another company, their frontier model is just so much better than the one that they are using right now.

Chris Hennesey (00:25:09)
Yes, I think the benefit of that singular frontier model is there's a simplicity and predictability that exists there. But I think as you've said, well, that nimbleness is going to be key. And I think that's a key element of this.

We all have seen one model won't trump all. The pace of innovation is pretty astounding across the frontier. So, the companies that are leveraging that singular approach, I think when I have conversations with them, while they've made agreements and they've aligned around a model type, they also are actively testing and keeping an active eye on the landscape that's out there and competitive landscape.

Because it's pretty clear for anybody close to this is that it's going to evolve quickly and there's a fungibility that's necessary here. If you even think about commitment horizons from cloud providers on models with customers, they're pretty short durations because the pace of change is so much. So, I have not seen companies who are so hardened in their ways that they won't be able to be fungible or flexible.

But because it's so early in the journey, they've pretty much are deciding to lean in with some single model providers, but having an active eye on how do things need to evolve in the event we need to pivot through time. So, I think it's a changing landscape. I think in just even listening to your podcast, it's interesting to hear the evolution, even in months, of how things evolve and the advice.

But still, a lot of the customers I meet with are leaning in with one single provider of models. That doesn't mean that they don't have the infrastructure flexibility to pivot, but they're seeing great results and they're actually doubling down on those results because they're just seeing the progress and the payback inside their organization.

Anurag Rana (00:26:49)
It's very interesting. I also came across a company today that's basically created its own proprietary LLM, and I was a bit surprised at that just because this is not a frontier lab, not a whole lot of money spent. Do you think that's a direction companies will take down the road? Because this way, they not only keep all the IP in-house and not worry about somebody using it to train their own models down the road, but that has to be a significant cost savings.

Chris Hennesey (00:27:18)
Yes. I think it's a balance because depending on how that model initial LLM was created, there is a lot of investment that's necessary unless they're kind of forking an open-source model and approach there. I think the balance between build versus buy versus kind of customize, kind of taking the RAG approach of buying the model and then improving the model with the data that exists in there and fine-tuning it. I have not seen the trend you mentioned where companies are building LLMs into that regard.

Where I have seen that be a consideration is when there's, one, huge data sets and large customer base that they can train from that, from an open weight model and improve it from there and take advantage of that. I don't see a lot of companies going that route. I'm not sure, given the pace of change of models, if it'll evolve that way too much in the near term, but I do think in the long term there could be the potential for small models being locally developed once there's a predominant kind of ecosystem that exists out there, and especially for high volume where there's enough volume to kind of reduce the unit cost enough in consuming that locally.

The things that, especially in Europe, this comes up a lot for me, where data sovereignty and regulations are very, very sensitive. That could be a really interesting approach that some companies could take, and I've seen some pharma companies looking at local LLM deployments in my engagements. I do think there could be an advantage, obviously, on the competitive side, but I think the pricing and value and cost side is something that they're going to have to consider, and I think that's how quickly some of the frontier model organizations are keeping a close eye on how they price products and services, because the economics would be hard to work to train your own models, and given the pace of change, it would be hard to do. That's not to say it won't evolve to that. I think it sounds like you found an example there, which is pretty good.

I think the hybrid approach could be one, where let's use a frontier model for these things. Let's fine tune a small model for these things that could be self-hosted. I think it's not going to be one or all. It could be all of them, and just a hybrid approach around that, but it's not coming up in a lot of the conversations I'm talking with customers.

Anurag Rana (00:29:40)
Fair point. Now, you talked about token-based stage when you first started. What are some of the ways companies can eliminate token-based stage in your experience right now?

Chris Hennesey (00:29:52)
Yes. You never want to pay twice for the same answer. So, I think repeated queries that are hitting models, a lot of companies are using both prompt and semantic level caching. So, how do we have a really good watchful eye and an awareness of all the queries that are happening on LLMs, and then how do we leverage a caching approach to that?

I think that's one definitely there. I think, as you know, the pricing of these models is based on both inputs and outputs. So, how much context do you give the model, and is it necessary to get the right output?

So, being really mindful of the inputs, but then also, how do you right-size the output? So, I've seen a lot of more customers putting cap responses in place, asking for a certain structured output as a part of this versus verbose outputs that potentially could come. So, I think that right-sizing the outputs and the inputs is the second lever that I see a lot from customers.

There's the saying, retrieve less, retrieve better, is something I've heard from customers, which is, how do you really be very deliberate about the knowledge base and the retrieval discipline that exists inside of the ecosystem that exists there? I think, obviously, model choice is a big deal. I think the number I've seen, the last number I saw was 150 times difference between an open-weight, open-source model and the highest-cost frontier model.

That's a large scale of difference that exists in there, and obviously, there's a lot of choice that's there. So, being deliberate about what model should be leveraged for what task, and that intelligent routing and having the systems do that would be kind of a fourth lever that I've seen customers kind of use to eliminate token wastage.

Anurag Rana (00:31:36)
We had an AI startup on the podcast a few months ago, and he was telling us that for repeated transactions, they just convert the discussion into a workflow, and in that case, you don't even have to use a token. I thought that was a very novel idea. Do you see that often anywhere?

Chris Hennesey (00:31:53)
Yes. I haven't seen that much, but I do see customers creating their own knowledge bases, and maybe that's the same spirit of what you're mentioning, which is, how do we -- because one of the talk tracks I have with customers is, how do you move from one times value to 10 times value? And one of the elements I'm seeing is there's a lot of federated local optimization, especially with [Agentic AI - (00:32:13], that's happening in isolation inside of different departments inside of enterprises, but there's a need to integrate that all together, and some of that could be done through knowledge bases and flows, as you talked about.

So, being deliberate about to see the 10 times, that's where you really want to ensure that you're really being deliberate about, how do you have shared data that you're leveraging across multiple applications? How do you then have reusable components as a part of that? So, while there may be some cost to deploying it on the front end, you're seeing so much scale as you go through more of a platform approach versus a vertical approach.

But I love the idea of the flows, but also, as you know, the knowledge base is key, and that's true for even the work I do in Amazon within our team. We also are trying to create our own knowledge base as a team, so we can also have the power of perspective of all, not just any one of us individually. So, I think it could evolve to those data stores or knowledge bases or flows through time, but again, it's so early in the journey.

Most are just trying to figure out what's the art of the possible for AI, for their jobs and the capabilities, and just trying to see what their strategy is and how they're going to deploy it inside their corporation.

Anurag Rana (00:33:27)
Yes. Another example somebody gave me that they were using the Qwen model by Alibaba for ERP-related operation where it was batch processing. It wasn't something you need –

Chris Hennesey (00:33:40)
Yes.

Anurag Rana (00:33:40)
-- in real time. And they said, well, the CFO's team does not need the answer right away. They can come back tomorrow morning and give me that information. So, that reduced their costs a lot. Is that something you've heard as well?

Chris Hennesey (00:33:54)
Yes. I think there's a lot of immediacy that people are looking for as they do the work, but I think the intentionality of batch processing is a really good approach. As a lever, I hear it talked about, but not as actively as I would expect in terms of using it. So, I think it is a lever that most have as an option and that exists.

And I think this is something that's going to evolve through time, because for almost every one of these levers that I talked about, it could result to a 40% to 50% reduction in costs, in inference costs specifically. So, compounding some of these together is pretty material. While batching is a clear strategy, most of my company, the companies that I engage with, most are still working in a real time way. So, I think that maturing is going to come through time.

And I know you're doing this podcast because the headlines over the last six months, a lot of it has been questions on, one, predictability, two, what's the cost profile? And then the third is a lot of things have been different than expected in terms of relative to budget. So, I think

the stat I just read last week was something like 69% of companies have spent more than their budget on AI this year. And I believe that was from Gartner.

So, I think that gives you a window into, one, companies need help and guidance on how to predict. But then, two, what are the levers you can pull? And batching is definitely another good lever out there.

Anurag Rana (00:35:22)
You said you talk to CFOs and you tell them about forecasting and ranges. So, let's say, hypothetically, if you come to, I'm running a company, you come over and say, well, let me give you how to even start thinking. So, give us a framework of, let's say, four or five points that how should one even go about it?

Chris Hennesey (00:35:43)
Yes. I think there's a couple of things to do. One, you've got to kind of bifurcate it to the components. And I always like to get into usage elements so it's more predictable.

So, one, who are the users and the use cases of individuals that will be leveraging it? So, getting a sense of the size of the magnitude. What level of workflows do you expect to leverage as a part of this? So, what's the size of the opportunity on the volumes that exist out there?

You then need to translate that in terms of, okay, great. Now that I have a sense of who's going to use it, how they intend to use it on a high level. And then you need to get into, great, how do things actually get priced with all the software vendors and the AI vendors that exist out there? So, you have to start building a muscle around the elements of token pricing and translating things into tokens. I know sometimes when you start talking about tokens, it's very hard for people to conceptualize. How do you even expect what that is?

But I think as you get into the specifics of this, you need to start breaking down some of the elements and core components that exist there. So, both input and output tokens, and then looking at ultimately what's the rate you're going to assume on that. That gets into what models you're going to leverage and what are the ways you're going to go through that.

So, most customers, as you start breaking apart the components, which is what are the models we're going to use? Who are the users are going to use it? What are the prompts they're likely to do? And then how do we get into an estimation of that?

So for me, it's just about breaking down the components into three variables. What's the costs in the token? How many tokens are you going to request? And then what are the requests per day? So, those three independent variables will enable you to predict.

The challenge people have is locally, they could maybe think about this, but at an enterprise scale, it becomes more challenging. The beauty is once you have data and usage patterns on this, you can start leveraging even AI to predict your expected outcomes as this. And just like

any good CFO, I always love to kind of evolve from point estimates to kind of a range of outcomes. So, you could think of a base high and low scenario as you go through this.

So, I think that keeping an eye on the unit economics is key, but also breaking it down to the component parts and understanding how all the pricing happens within the model providers itself.

So, I'm sure I don't know all the platforms you leverage. When you're in a subscription view, it's pretty straightforward, you can sum it up and you're good. And especially for companies using, say, Copilot, the licensing is clear.

Bedrock for AWS, the usage of that is clear, but it's the variable scaled side of the usage patterns as where the unknowns are. So, I just like to forecast and plan it out and put estimates on it and then compare to your accuracy through time.

And you can continue to hone this as it scales inside of your organizations. And especially as things get more automated, there's a lot more predictability that happens as you get into the cost forecasting side.

Anurag Rana (00:38:45)
Fair point. So, the cost forecasting one side, now comes the, I guess, the toughest part of it is what framework do I use to do even an ROI measurement for AI investments that I'm doing? So, you spending all that token, what is really my return? How to come up with that framework?

Chris Hennesey (00:39:03)
Yes, there's a couple of things to consider. And there's like two approaches I've seen companies use the most. One is they take almost like a venture capital kind of mindset, which is I don't want to estimate every use case, but I do know collectively we need this to pay off.

So, they won't get too bogged down in any one use case and the estimation of it. But they will be deliberate about looking at the portfolio approach and ensuring that they're seeing a returns at that aggregate level. And there's the saying, one out of 10 companies is successful, so the goal is then start 10 companies because one will be successful.

That's that portfolio approach or VC approach I've seen with many companies, which is we're going to do a lot of initiatives and efforts around AI. Some of these are going to deliver outsized returns. Some of these are going to fail. And some of these we need to keep a close eye on.

So, they take that portfolio ROI view. It's not a project-based portfolio view. So, I think there's a suite of companies that are there. Some customers aren't comfortable not quantifying all the specifics.

So, they focus a bit more on the outcome-based metric approach, which is instead of

measuring AI directly, they're measuring what's the intended outcome of that. And it's really through a myriad, and I know the general kind of advice I give to customers is you got to move from kind of leading indicators and metrics to lagging as you go through this. Because the first beginning of this, and I'm sure this is true in most of the customers you talk to, is there's a lot of activity happening on the front end.

And this would be a bit about like what's the adoption rate of AI? How much is co-development AI developed versus not? I heard that on some of the examples in your other prior podcasts.

So, you evolve from activity to ultimately, customers usually evolve into more of the quality side. So, think about like CSAT on interactions for customers, or are you resolving things? It eventually evolves into efficiency metrics, which a lot of customers are focusing on, which is how is this changing the cost per whatever that outcome may be.

And then eventually it's hitting the P&L, which is the business outcome through time. But it takes a bit of time for that to cure. You're not going to see it.

So, I think the leading and lagging kind of metric stack is something that I talk to customers a lot about on the AI journey as they think about measuring returns. Because just because you're using a coding assistant, that doesn't mean you're resulting in more revenue or less losses for the company. You've got to have measurements that assess throughout that whole cycle.

Anurag Rana (00:41:33)
Yes, I was just coming to that. Because see, you just talked about a particular statistic, which is S&P growing 8%, but expenses growing 14%, right? At the end of the day, capitalism, it has to come up and show up somewhere.

I mean, it's not a -- this is not all fun and games over here. But even when we look at something that is absolutely massive in the last 12 months, which is coding you listen to any podcast out there, every new startup is saying 75%, 80%, 90% of the code is now generated by AI. And yet, we have not seen any kind of massive layoffs with software developers.

Now, maybe there is enough tech debt out there that we will see. But it has to come on one of two ways, whether we're going to see the margins improve or the revenue go up. Where do you think the -- where are we going to get the money to pay for all of this?

But, I'm only talking about Fortune 2000 companies. Let's not even go how the frontier models are funding themselves. Let's just keep that on the side.

Chris Hennesey (00:42:38)
Yes. Yes, I would agree with your assessment. I know most of the organizations I'm engaging with, large enterprises, they're reabsorbing or reinvesting a lot of that incremental capacity and productivity gains.

I think there's probably, again, if you were to split it, three-fourths of them who are locally reinvesting it into where that productivity gains are. But I've started to see some companies redistribute capacity inside of their organization based on these efficiency gains. So, while they may not be bringing down the water level of labor at this point, at least internal labor, they are redirecting that labor to where the most pressing needs are inside of the organization.

One of the trends I'm seeing, and I've heard it literally in the last four weeks from customers, is there's a lot of efficiencies being gained on offshoring and third-party labor engagements. So, they're starting to see the efficiencies. They want to retain the talent, especially on the development side, for investment capacity needs in the future.

But they're willing to offload some third-party resources as a part of this. So, I think that reinvestment is a key decision point and framing because that reinvestment is ultimately going to result in more value and revenue and capabilities for the organization. So, new products, new services, new capabilities that exist there.

So, I'm also not seeing the layoff kind of headlines. AI may be contributing to some of them, but in reality, it's not to the scale of the investment. So, I think that it's all about kind of growing the share of the company and the products that they have and expanding in this regard.

So, that metric is one I'm going to keep a really close eye on because the other thing I'm seeing in the stats I gave you was IT costs compared to revenue. The other thing that's happening is IT cost is growing at a fast rate. But other costs inside the company is actually declining or staying flat.

So, there's a composition change happening in a lot of companies where more software is being procured, but that could be it's growing at a faster rate than cost outside of that, especially in operations or other functions. So, cost composition is something I'm keeping an active eye on. The other thing is the return profile through time.

So, I think, again, early in the journey, there's investments being made. Payback is expected. When that payback occurs, that horizon is very different for every different vertical that I've seen.

But it's something to keep an active eye on as we go through this because, to your point, [math - 00:45:03], economics needs to work. You can't grow fat costs at a higher rate of revenue over the long term. But I think some of this is IT cost is growing faster. The rest of the company is not growing as much, if not declining, in some cases where automation is occurring.

Anurag Rana (00:45:18)
It's a very good point you say. And again, I'll tell you, one of the biggest human-centric

industries is tech services, whether it's the offshore vendors or it's just --

Chris Hennesey (00:45:27)
Yes.

Anurag Rana (00:45:27)
-- consulting companies. At the same time, I mean, custom app development, deployment of software, if the cycle times are going down, I'm just surprised.

And again, none of these vendors are growing right now because of budget constraints on the on Fortune 2000 side. But I'm still surprised they're still hiring almost on every quarter. And it just doesn't make sense to me.

What do you think is really happening over there in terms of do they have extra work while deflation is happening? Or they're just hoping that they will get more AI-related work to deploy down the road?

Chris Hennesey (00:46:04)
Yes, they get -- I think it's a blend. I think one, you need to -- the other thing factor that I would consider in that is there's kind of a talent dynamic that's going on now where there's a strong desire and a need for not only developing kind of your internal skill, your internal talent with AI skills, but also there's a high attractive element of attrition that's occurring in some of the customers I'm engaging where the AI experts are being attracted to go elsewhere.

So, you also have to substitute and backfill some of that talent that exists there. But I think it's a blend and a balance here. I think the growth opportunities exist.

I think we're also just in a large investment cycle as you think about the technology that's here. And there's a belief that there's going to be large payback and capability that exists here. And any one of us who leverage AI tools pretty actively can see that this is opening up more capabilities.

It's enabling us to reach customers faster and in new ways. I know for me, I meet with a couple hundred customers a year. For every customer engagement now, I'm creating kind of vibe coded platforms to demo the capabilities of leveraging agentic AI inside their workforce.

Well, that won't scale, especially vibe coding won't scale. You need great technologists to enable that. It's the art of the possible around how this could transform the way in which people do work and the way in which they could take away, especially a lot of manual effort across a lot of enterprises.

How can you automate and augment a lot of those experts with more automation and capability? So, I think the size of the pie is increasing, to your point, of where the art of the

possible is and the capabilities within enterprises.

Anurag Rana (00:47:48)
Great. Chris, I think we are almost over time. In fact, I'm very certain we are over time.

Chris Hennesey (00:47:53)
Yes.

Anurag Rana (00:47:53)
But let's just do one last question before we close. Any parallels to the initial growth of AWS or the formation of AWS when the cloud shift happened to the AI journey now?

Chris Hennesey (00:48:08)
Yes. I think at least from the cost and value side, there are definitely a lot of parallels. One, I know when I was going through a cloud transformation, it took a bit of time to build the muscle internally to know how to use it well. Bringing an on-prem mindset to the cloud, that ultimately resulted in a lot of cost overrun because you weren't taking advantage of the elasticity and scalability.

So, I think there was a whole FinOps discipline that had to be built inside of organizations. I think two, to the parallels here, just because you're the biggest spender doesn't mean you're going to deliver the biggest value. So, I think, one, I've seen the creation and the new startups being born every week. And it's no easier than ever now to build those capabilities.

And then two, the last thing is the technology is never the constraint, specifically. One of the biggest themes and talk tracks that it's most engaging for our organization is around operating model and culture that we've had. There's a book called The Octopus Organization written by Jana Werner and Phil Le-Brun, two members of the team I'm on. I've never seen something kind of grow as much in interest from customers because a lot of the constraints are not the architecture and the LLMs and the models. It's more of how do I take advantage of that capability?

And I think that was true for the cloud as well. It started to need to break down the capabilities and what the art of the possible is. If I didn't have to invest capacity into this, what else would I do with that capacity? And that we're in that same inflection point with AI, too.

Anurag Rana (00:49:41)
Great, Chris. This has been so much fun. Thank you so much for your time today.

Chris Hennesey (00:49:45)
I appreciate it. Thanks so much for having me.